MIL-B-1995E
24 May 1978
SUPERSEDING
MIL-B-1995D
15 June 1972

MILITARY SPECIFICATION

BRUSH, GREASE, COOKING AND BAKING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of grease brush.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

[illegible]-S-951	-	Synthetic Filaments for Brushes, General Specification for
TT-B-[illegible]	-	Butyl Acetate, Normal (for use in Organic Coatings)
TT-B-[illegible]	-	Butyl Alcohol, Normal (Butanol) (for use in Organic Coatings)
TT-M-261	-	Methyl Ethyl Ketone, Technical
TT-T-548	-	Toluene, Technical
UU-P-268	-	Paper, Kraft, Wrapping
PPP-T-45	-	Tape, Gummed, Paper, Reinforced and Blade, for Sealing and Securing
PPP-T-76	-	Tape, Packaging, Paper (For Carton Sealing)
PPP-B-566	-	Boxes, Folding, Paperboard
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-B-676	-	Boxes, Set-up

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research and Development Command, Natick, MA 01760, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document of by letter.

FSC 7330

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Units Loads for 40" x 48" Pallets
MIL-STD-1363 - Measurement of Wood Moisture Content

(Copies of specifications, standards, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

LAWS AND REGULATIONS

21 CFR 177 - Subpart B - Substances for Use as Basic Components of Single and Repeated Use Food Contact Surfaces

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, US Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for the issuance thereof.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

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(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Standard product. The grease brush delivered under this specification shall be the manufacturer's current standard commercial product, except for any changes necessary to comply with specification requirements.

3.2 Materials. The hog bristles or synthetic filaments and materials not definitely specified shall be of the quality normally used by the manufacturer for brushes provided the completed items comply with all provisions of this specification. Commercial tolerances are applicable to all metal thickness and diameters (see 6.3).

3.2.1 Synthetic filaments. Materials used in the production of synthetic filaments and filler setting for the manufacture of the grease brush shall conform to the requirements of Subpart B (Substances for Use as Basic Components of Single and Repeated Use Food Contact Surfaces) or Subpart C (Substances for Use Only as Components of Articles Intended for Repeated Use) Part 177, Chapter I, Title 21, Code of Federal Regulations. The filaments shall conform to the physical characteristics of class N, type I, size 12-8 of H-S-951 except that the color is not restricted to black only.

3.2.2 Epoxy resin. Epoxy resin used for the setting of hog bristles or synthetic filaments shall conform to the food and drug requirements specified in 3.2.1.

3.2.3 Hog bristles. The hog bristles shall be sterilized. Hog bristles, that have not been previously sterilized, shall be sterilized prior to assembly after thorough washing and rinsing in accordance with any of the following methods: (1) soak bristles in boiling water, 212°F (100°C) for not less than 30 minutes; (2) place bristles in a dry chamber at least 284°F (140°C) for not less than 3 hours or, (3) place bristles in a steam pressure chamber at a minimum 15 pounds per square inch (103 kPa) for a minimum of 45 minutes.

3.3 Construction.

3.3.1 Filler material. The brush shall have natural hog bristles or synthetic filaments. Minimum total weight of the hog bristles or synthetic filaments for each brush shall be 1-1/8 ounces (32 g).

3.3.2 Handle. The handle shall be fabricated from beech, birch, cherry, maple, pecan, or sweetgum. The wood shall be dried to an average moisture content of between 8 to 12 percent at time of fabrication (see 4.2.1.1).

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3.3.3 Ferrule. The ferrule shall be steel or plastic. Steel ferrule, when applicable, shall be 0.018 inch, plus or minus 0.002 inch thick (0.46 ± 0.05 mm), and shall be nickel plated, with end lap not less than 1/4 inch (6.4 mm). End lap shall be either lock-seamed or securely spot welded. Plastic ferrule shall be plastic tube made of paper and phenolic resin and shall have a minimum thickness of 0.0625 inch (1.6 mm).

3.3.4 Brush assembly. The butt end of the hog bristles or synthetic filaments shall be set in a synthetic resin of the phenol-formaldehyde group or in an epoxy resin. Metal ferrule, when applicable, shall be placed around the hog bristles or synthetic filaments and handle. The metal ferrule shall be secured to the handle by at least four stainless steel or nickel-copper alloy nails or escutcheon pins; two to be driven from each side. The hog bristles or synthetic filaments shall be secured to the metal ferrule by at least two staples of sufficient length to permit clinching of the staples or may be set in an epoxy resin and a ferrule with a groove at the base of the setting compound. The epoxy resin shall not soften or deteriorate and the bristles or filament shall not loosen when tested as specified in 4.3.1. The plastic ferrule, when applicable, shall be assembled to the bristles or filaments and the handle by means of a synthetic resin, a phenol-formaldehyde resin, or other suitable resins. The end of the hog bristles or synthetic filaments on the finished brush shall have machine chisel ends.

* 3.3.5 Dimensional requirements. Dimensional requirements shall conform to the following:

TABLE I. Dimensions

Item	Inches	(mm)
Length handle above ferrule	5 ± 1/4	(127 ± 6.4)
Depth of ferrule	1-3/4 ± 1/32	(44 ± 0.8)
Width inside ferrule	3 ± 1/32	(76 ± 0.8)
Thickness inside ferrule	1/2 ± 1/32	(13 ± 0.8)
Length bristles or filaments inside ferrule	3/8 ± 1/8	(9.5 ± 3.2)
Length bristles or filaments outside ferrule	2-3/8 ± 1/8	(60 ± 3.2)
Overall length of brush	9, +1/2 , -1/4	(229, +13, -6)

3.4 Marking for identification. The brush shall have the manufacturer's name, trade name, or trademark readily identifiable with the manufacturer, embossed or stamped on the ferrule or handle.

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3.5 Finish. The handle shall be finished by applying transparent varnish or lacquer.

3.6 Workmanship. There shall be no sharp edges, splinters, burrs, slivers, or roughness which may cause personal injury. Staples, nails and pins shall be firmly set. Handle shall be uniformly covered. Filler must be firmly set in ferrule. The brushing part shall be trimmed evenly and uniformly to the finished length outside the ferrule.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certification of compliance. The contractor shall make available to the Government representative proof of compliance with Subpart B (Substances for Use as Basic Components of Single and Repeated Use Food Contact Surfaces) or Subpart C (Substances for Use Only as Components of Articles Intended for Repeated Use) Part 177, Chapter I, Title 21, Code of Federal Regulations (see 3.2.1) and [illegible] the requirement for sterilization of hog bristles (see 3.2.3).

4.2 Inspection. Except where otherwise indicated hereinafter, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.2.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification.

4.2.1.1 Moisture content test for lumber. A moisture content test shall be performed on each lot of lumber offered for inspection at time of fabrication in accordance with MIL-STD-1363. The sample unit shall be one piece of lumber. Three determinations shall be taken on each sample unit and results shall be the average of the three determinations (see 3.3.2). The inspection level shall be S-2 with an acceptable quality level (AQL) of 2.5, expressed in terms of defects per hundred units.

4.2.2 Intermediate inspection.

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4.2.2.1 Hog bristles or synthetic filaments. Inspection shall be made of the hog bristles or synthetic filaments to determine conformance with weight requirements of 3.3.1. The inspection lot shall be the amount of brush filaments or bristles for attachment to ferrules offered for inspection at one time. The sample unit shall be one assembly of brush filaments or bristles. The inspection level shall be S-2 with an AQL of 2.5 expressed in terms of defects per hundred units.

4.2.2.2 Dimensional inspection. Examination shall be made of the length of bristle or synthetic filaments inside the ferrule to determine that the dimensional requirements of 3.3.5 are complied with. The inspection lot shall be the number of brush assemblies offered for inspection at one time. The sample unit shall be one bristle or synthetic filament assembly. The inspection level shall be S-2 with an AQL of 4.0 expressed in terms of defects per hundred units.

4.2.2.3 In-process examination. Examination shall be made during the manufacturing process listed below. When nonconformance is noted, correction shall be made to the affected items and process. Sterilization of hog bristles is in accordance with 3.2.3.

4.2.3 End item inspection. The inspection lot shall consist of all brushes offered for inspection at one time. The sample unit for this inspection shall be one completely fabricated brush.

4.2.3.1 Visual examination. The end item shall be examined for defects classified in table II. The inspection level shall be level II with an acceptable quality level (AQL) of 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

TABLE II. Classification of defects

Examine	Defect	Classification Major	Minor
Finish	Handle splintered or rough.	X	
	Metal ferrule not plated.	X	
	Varnish or lacquer not transparent.		X
	Handle not covered or poorly covered with varnish or lacquer.	X	
Construction and workmanship, (general)	Any part missing (except when otherwise classified herein).	X	
	Burrs, rough edges, or slivers on metal parts.	X	

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TABLE II. Classification of defects (cont'd)

Examine	Defect	Classification Major	Minor
Brush assembly	Setting compound (synthetic resin) beyond ferrule.		X
	Metal ferrule (not grooved) when set in epoxy.	X	
	Metal ferrule (when applicable) not secured to handle by at least four nails or escutcheon pins, two driven from each side.		X
	Bristles or filaments not secured to metal ferrule (when applicable) by at least two staples.	X	
	Staples (when applicable) not of sufficient length to permit clinching.	X	
	Metal ferrule (when applicable) not lock-seamed or lap jointed and spot welded.	X	
	Loose fit of ferrule to bristle or filament assembly.	X	
	Bristles or filaments loose in the setting compound.	X	
	Loose fit of ferrule to handle.	X	
	Bristles or filament not finished with chisel ends.		X
Marking for identification	Missing, incomplete, not legible, not in proper location, not stamped or embossed.		X

4.2.3.2 Dimensional examination. Examination shall be made of the end item to determine compliance with dimensions specified. Any noncompliance with specified requirements shall constitute a defect. The inspection level shall be S-2 with an AQL of 4.0 expressed in terms of defects per hundred units.

4.2.3.3 Testing of the end item. The end item shall be tested in accordance with 4.3.1. The inspection level shall be S-2 with an AQL of 4.0 defects expressed in terms of defects per hundred units.

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4.2.4 Packaging inspection. An inspection shall be made to determine that preservation-packaging, packing, and marking comply with section 5 requirements. Defects shall be scored in accordance with defects set forth in table III. The sample unit shall be one shipping container fully prepared for delivery except it need not be closed. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 with an AQL of 4.0 expressed in terms of defects per hundred units.

TABLE III. Inspection of packaging

Examine	Defect
Marking (exterior and interior container)	Missing; incorrect; illegible; of improper size, location, sequence, or method of application.
Materials	Component missing, damaged, or not as specified. Absence of proper amounts of napthalene balls or napthalene flakes (nog bristles).
Workmanship	Closure of intermediate package not accomplished or not correct material.
Contents	Number per container is more or less than required.

4.2.4.1 Inspection of shipping containers. When shipping containers are required to be in accordance with PPP-B-636, examination for defects in closure, waterproofing, and strapping shall be in accordance with the appendix of PPP-B-636.

4.3 Tests.

4.3.1 Solvent resistance. The end item shall be tested for solvent resistance using a solvent composed of the following mixture:

	Percent by volume	Specification
Normal butyl acetate	15 ± 1	TT-B-838
Normal butyl alcohol	15 ± 1	TT-B-846
Methyl ethyl ketone	10 ± 1	TT-M-261
Toluene	Remainder	TT-T-548

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The brush shall be suspended so that the brushing part and approximately one-half of the ferrule are immersed in the solvent mixture for 1 hour at a temperature of 21° to 32°C. The brush shall be removed and examined as follows.

4.3.1.1 Ferrule. As soon as the brush is removed from the solvent the ferrule shall be examined for wrinkling, blistering, separation of the coating or perceptible change in material. Fillers or strips shall be examined for softening or perceptible change in material. The brush shall then be air-dried for 2 hours and the coating examined for softening, dulling, or discoloration. Any nonconformance shall constitute failure of this test.

4.3.1.2 Setting compound. At the time the brush is removed from the solvent, the setting compound shall be examined for excessive softening, solubility in the solvent, or loosening of the bristle or filament. The brush shall then be air-dried for a period of 24 hours, the ferrule removed, and the setting compound reexamined for evidence of deterioration or loosening of the bristle or filament. Any nonconformance shall constitute failure of this test.

5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unit package. Each brush shall be individually wrapped in minimum 30-pound (49 g/m^2) kraft paper conforming to type I, grade B of UU-P-268. The ends of the kraft paper shall be folded back over each end of the brush and secured with one inch (25 mm) minimum width gummed paper tape conforming to type III, grade B of PPP-T-45. Alternatively, the bristle end or filament end (including the ferrule) of the brush shall be encased in a wallet-type envelope of kraft paper as specified herein, or polyethylene envelope or sleeve. The envelope shall be secured to the handle of the brush by attached tie strings or tape as specified herein. A rubber band minimum 1/8 inch (3.2 mm) wide, may be used in either method instead of tape or tie strings.

5.1.1.2 Intermediate package. Twelve brushes, unit packaged as specified in 5.1.1.1, shall be packaged in a snug-fitting folding paperboard, set-up paperboard, or fiberboard box conforming to variety 1, style III, type G, class 1 or style IV, type I or J, class 1 of PPP-B-566; or type I, variety 1, class A, style 2 of PPP-B-676; or style RSC, type CF, variety SW, or type SF, class domestic of PPP-B-636, respectively. Each box containing brushes with hog bristles shall contain a minimum of three balls of napthalene or an equivalent quantity of flakes. Boxes shall be closed by covering all

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seams or openings full length with strips of 2-1/2 inch (64 mm) minimum width tape conforming to type III, grade B of PPP-T-45 or PPP-T-76. For fiberboard boxes, the center strip shall extend a minimum of 2-1/2 inches (64 mm) onto the end panels.

* 5.1.2 Level C. The brushes shall be packaged to afford adequate protection against damage during shipment from the supply source to the first receiving activity. The package and the quantity per package shall be the same as that normally used by the contractor for retail distribution.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. One hundred forty-four brushes, preserved-packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, grade V2s of PPP-B-636. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced as specified in the appendix of PPP-B-636.

5.2.2 Level B. One hundred forty-four brushes, preserved-packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, type CF, variety SW, or type SF, class domestic of PPP-B-636. Each shipping container shall be closed in accordance with method II of the appendix of PPP-B-636.

5.2.2.1 Weather-resistant shipping container. When specified (see 6.2), the fiberboard shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of PPP-B-636.

5.2.3 Level C. Brushes, preserved-packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with Uniform Freight Classification or National Motor Freight Classification, as applicable.

5.3 Palletization. When specified (see 6.2), brushes packed as specified shall be palletized in accordance with Load Type I of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L. Pallet patterns shall be in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the pallet patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contracting officer.

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5.4 Marking. In addition to any special marking required by the contract or order, intermediate packs, shipping containers and palletized unit loads shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The brush covered by this specification is intended for coating baking pans, cooking pans, and utensils with oil or shortening and for applying a wash or glaze to bakery products such as bread or rolls.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Selection of applicable levels of preservation-packaging and packing (see 5.1 and 5.2).
(c) When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).
(d) When palletization is required (see 5.3).

* 6.3 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification (see 3.2).

* 6.4 Metric equivalents. Metric equivalents, indicated in parentheses throughout this document, are based on practices, conversion factors, and symbols specified in ASTM E 380 Standard for Metric Practice, and are for information only. In each instance, the value stated in US customary units shall be controlling. Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.

6.5 Changes from previous issue. The outside margins of this document have been marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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DOCUMENT IDENTIFIER AND TITLE: MIL-B-1995E
BRUSH, CREASE, COOKING AND BAKING

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

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   A. GIVE PARAGRAPH NUMBER AND WORDING

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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two or more stitches less than minimum